

Does the Order of the Terms Matter?

Rearrangement of Infinite Series

Engineering Mathematics–I (CSE-1001)

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Kanchan Rajwar

kanchan.rajwar@iiitbhopal.ac.in

This is a short supplementary note, written in answer to a question asked in class:

“Rearranging the terms of a conditionally convergent series cannot change its sum — true or false?”

The answer is **false**, and the reason is one of the most surprising results in the whole of first-year analysis. Read Section 11 of the companion note *Series* (Absolute and Conditional Convergence) before this one; everything here builds on that section’s definitions.

1 The Question

Addition of finitely many numbers does not care about order. For any finite list,

$$2 + 7 + 5 = 5 + 2 + 7 = 7 + 5 + 2 = 14.$$

This is the *commutative law*, and we use it so automatically that we rarely name it. It is natural to expect the same of an infinite sum.

It is not true. For some infinite series the order of the terms is part of the data, and shuffling them changes the answer.

The idea

A finite sum is a single completed operation. An infinite series is a *limit of partial sums*. Rearranging the terms produces a *different sequence* of partial sums — and different sequences may perfectly well have different limits. The commutative law was never promised for limits.

2 A Counterexample

Recall from the *Series* note that the alternating harmonic series converges conditionally:

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \ln 2 \approx 0.693147. \quad (1)$$

It converges by Leibniz's test, but the series of absolute values is the harmonic series $\sum \frac{1}{n}$, which diverges. That is exactly what *conditional* convergence means.

Now reorder the very same terms, taking **two positive terms, then one negative**, and repeating this pattern for ever:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots \quad (2)$$

Every term of (1) appears in (2) exactly once. Nothing has been inserted, nothing deleted, nothing repeated. Only the order has changed. Yet

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots = \frac{3}{2} \ln 2 \approx 1.039721.$$

The sum is one and a half times what it was.

2.1 Proof that the new sum is $\frac{3}{2} \ln 2$

This takes three lines. Write out S , and underneath it write $\frac{1}{2}S$ with a zero inserted before each of its terms so that the columns line up:

$$\begin{array}{rcccccccc} S & = & 1 & -\frac{1}{2} & +\frac{1}{3} & -\frac{1}{4} & +\frac{1}{5} & -\frac{1}{6} & +\frac{1}{7} & \cdots \\ \frac{1}{2}S & = & 0 & +\frac{1}{2} & +0 & -\frac{1}{4} & +0 & +\frac{1}{6} & +0 & \cdots \\ \hline \frac{3}{2}S & = & 1 & +0 & +\frac{1}{3} & -\frac{1}{2} & +\frac{1}{5} & +0 & +\frac{1}{7} & \cdots \end{array}$$

Deleting the zeros from the bottom row leaves precisely the series (2). Hence that series sums to $\frac{3}{2}S = \frac{3}{2} \ln 2$. ■

Why inserting zeros is legitimate

Adding a zero before each term of a convergent series changes neither its convergence nor its sum: the new sequence of partial sums is the old one with each value repeated once, so it has the same limit. Deleting zeros is the same operation read backwards. What we are *not* allowed to do casually is the rearrangement itself — and the whole point of this note is that we did not do one here. We added two convergent series term by term, which is always valid.

2.2 Visualization

Figure 1 plots the partial sums of the two series side by side. Both oscillate and both settle down — but to different places.

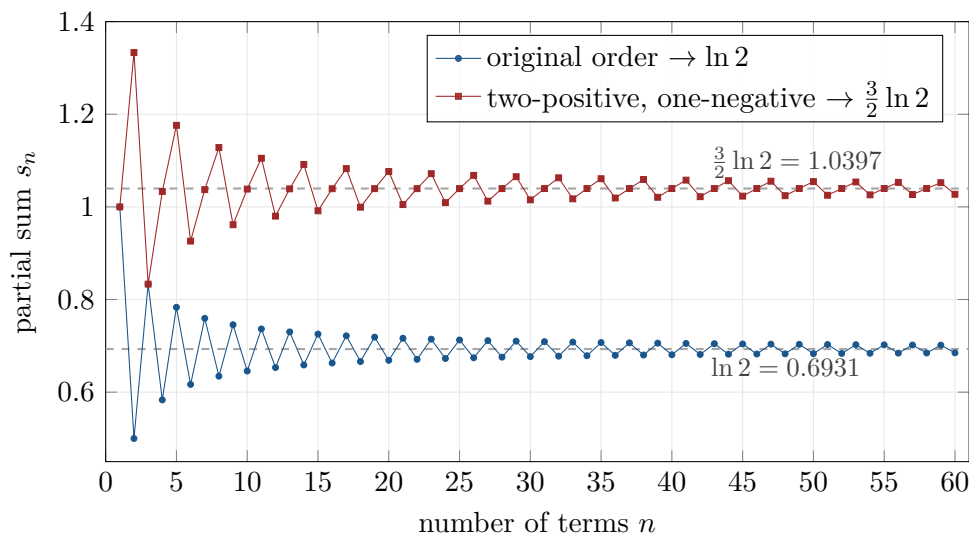


Figure 1: The same terms, added in two different orders. The partial sums converge to two different numbers.

3 Riemann's Rearrangement Theorem

The counterexample is not an isolated curiosity. It is the mildest case of a far stronger statement.

Theorem 3.1 (Riemann's Rearrangement Theorem, 1854). *Let $\sum a_n$ be a **conditionally** convergent series of real numbers, and let T be any real number. Then the terms of $\sum a_n$ can be rearranged so that the rearranged series converges to T . The terms may also be rearranged so as to diverge to $+\infty$, or to $-\infty$, or to oscillate without any limit at all.*

Read that again. Not “the sum might change”, but: *the sum can be made to be whatever you wish*. The series $1 - \frac{1}{2} + \frac{1}{3} - \dots$ can be reordered to sum to 7, to $-\pi$, to 10^6 , or to nothing at all.

The proof is a construction, and the construction is easy to follow. Separate the terms into the positive ones $p_1, p_2, p_3, \dots$ and the negative ones $-q_1, -q_2, -q_3, \dots$ (with each $q_k > 0$). Two facts drive everything.

Proposition 3.2. *If $\sum a_n$ converges conditionally, then*

$$\sum_k p_k = +\infty \quad \text{and} \quad \sum_k q_k = +\infty,$$

while $a_n \rightarrow 0$.

Proof. Suppose both $\sum p_k$ and $\sum q_k$ were finite. Then $\sum |a_n| = \sum p_k + \sum q_k$ would be finite too, making the series absolutely convergent — contrary to hypothesis. So at least one of them is infinite. Suppose exactly one is, say $\sum p_k = \infty$ while $\sum q_k$ is finite. The partial sums of $\sum a_n$ are then of the form (a quantity tending to $+\infty$) minus (a bounded quantity), which tends to $+\infty$ — contradicting convergence. Hence both are infinite. Finally $a_n \rightarrow 0$ because $\sum a_n$ converges, by the n -th term test. $\square$

The mechanism in one sentence

A conditionally convergent series carries an *unlimited supply* of positive material and an *unlimited supply* of negative material, in packets that become arbitrarily small — so you can steer the running total anywhere you like, and then park it there.

The construction. Fix a target T . Then:

1. Take positive terms, in order, until the running total *first* exceeds T . This must happen, because $\sum p_k = \infty$.
2. Now take negative terms, in order, until the running total *first* drops below T . This must happen, because $\sum q_k = \infty$.
3. Return to step 1 and repeat, for ever.

Every term is used exactly once, so this is a genuine rearrangement. At each turning point the running total overshoots T by no more than the size of the last term used — and those terms tend to 0 by Proposition 3.2. So the partial sums are squeezed onto T , and the rearranged series converges to T . ■

A numerical illustration

Chasing the target $T = 2$ with the alternating harmonic series by exactly this recipe, the partial sums at successive turning points run

$$2.0001, 1.9915, 2.0003, 1.9918, 2.0005, 1.9922, \dots$$

straddling 2 with a band that narrows as the terms shrink. To make the series diverge to $+\infty$ instead, simply raise the target at each stage — aim for 1, then 2, then 3, and so on.

4 The role of Absolute Convergence

The contrast could not be sharper.

Theorem 4.1 (Rearrangement is harmless when convergence is absolute). *If $\sum a_n$ converges absolutely, then every rearrangement of it converges, and every rearrangement has the same sum.*

Sketch. Write $\sum a_n = \sum p_k - \sum q_k$. Absolute convergence makes both $\sum p_k$ and $\sum q_k$ finite. A series of non-negative terms has increasing partial sums, so its sum is the supremum of its partial sums — and a supremum is taken over a *set* of numbers, which has no notion of order. Rearranging therefore leaves each of $\sum p_k$ and $\sum q_k$ untouched, and hence leaves their difference untouched. □

	Absolutely convergent	Conditionally convergent
$\sum a_n $	converges	diverges
Positive part $\sum p_k$	finite	$+\infty$
Negative part $\sum q_k$	finite	$+\infty$
Rearrangement changes the sum?	never	to anything at all
Behaves like a finite sum?	yes	no

Why we bother with the distinction

Students often ask why the course separates “absolutely convergent” from “convergent” when both, after all, converge. Theorem 4.1 is the answer. Absolute convergence is not a technicality — it is the **licence to treat an infinite sum like a finite one**. Without it, an infinite sum is not quite a number: it is a number *together with* an order of assembly.

The rule to carry into the examination

A conditionally convergent series may **not** be rearranged, regrouped, or multiplied out term by term. The order of its terms is part of the data. Every one of those operations is safe for an absolutely convergent series, and unsafe otherwise.

5 A Remark on Finite Sums

It is worth saying plainly what has, and has not, broken down. The commutative law of addition is perfectly true, and nothing in this note contradicts it. What has happened is this: an infinite series is *defined* as

$$\sum_{n=1}^{\infty} a_n = \lim_{N \rightarrow \infty} s_N, \quad s_N = a_1 + a_2 + \cdots + a_N,$$

and each s_N is a finite sum, to which commutativity applies in full. Rearranging the terms does not disturb any individual s_N ; it produces an *entirely different sequence* $s'_1, s'_2, s'_3, \dots$ of partial sums. Two different sequences are entitled to two different limits, and that is all that occurs in Figure 1.

Why this matters beyond the classroom

Numerical software adds floating-point numbers in whatever order the loop happens to run, and the order affects the answer — which is why summation in scientific computing is done with algorithms (Kahan summation, pairwise summation) that control the error rather than with a naive loop. The mathematical phenomenon in this note is the extreme form of the same issue: when cancellation between large positive and large negative contributions is doing the work, the order in which you add is not a detail.

6 Try It Yourself

Try it yourself

1. Classify each series as absolutely convergent, conditionally convergent, or divergent:

$$(i) \sum \frac{(-1)^{n+1}}{n^2} \quad (ii) \sum \frac{(-1)^{n+1}}{\sqrt{n}} \quad (iii) \sum \frac{(-1)^{n+1}n}{n+1}$$

For which of them is rearrangement dangerous?

2. Show that

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots = \frac{1}{2} \ln 2,$$

a rearrangement of the alternating harmonic series taking one positive term and then two negative ones. *Hint:* group the terms in threes and simplify each bracket.

3. Explain in two sentences why the recipe of Section 3.1 cannot be carried out on $\sum \frac{(-1)^{n+1}}{n^2}$.

Answers: (1) (i) absolutely convergent; (ii) conditionally convergent; (iii) divergent, since the terms do not tend to 0. Rearrangement is dangerous only for (ii). (2) Grouping in threes gives $\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \cdots = \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \cdots\right) = \frac{1}{2} \ln 2$. (3) That series converges absolutely, so its positive and negative parts are each *finite*; the construction needs both to be infinite, and so has nothing to draw on.